

Chain-of-Thought Prompting: Eliciting Reasoning in LLMs

FAMAF - UNC

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eXplainable
Artificial Intelligence

Disclaimer ---

This course is based on

Explainable Artificial Intelligence

From Simple Predictors to Complex Generative Models

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<https://interpretable-ml-class.github.io/>

🍲 Chain-of-Thought Prompting Elicits Reasoning in Large Language Models ---

(Wei, Wang, et al. 2022)

Introduction & Related Work ---

🍷 Challenges in LLMs ---

- Scaling up model size alone has not proved sufficient for achieving high performance on challenging tasks such as arithmetic, commonsense, and symbolic reasoning.
- Large language models still have limitations in their ability to reason and understand the context of a situation.

Previous Work ---

- Models can generate intermediate natural language steps by training from scratch (Ling et al. 2017) or finetuning a pretrained model (Cobbe et al. 2021).
 - ▶ Costly to create large sets of high-quality rationales.
- Large language models offer the prospect of in-context few-shot learning via prompting (Brown et al. 2020).
 - ▶ Works poorly on tasks that require reasoning abilities.
 - ▶ Often does not improve substantially with increasing model scale.

Contribution ---

- Explores few-shot prompting for reasoning tasks using prompts structured as triples: $\langle \text{input}, \textit{chainofthought}, \text{output} \rangle$.
 - ▶ **Chain-of-thought**: a series of intermediate natural language reasoning steps that lead to the final output.
- Presents empirical evaluations on arithmetic, commonsense, and symbolic reasoning benchmarks, showing that chain-of-thought prompting outperforms standard prompting.

(Wei, Wang, et al. 2022)

Methodology ---

🍲 How Do We Solve Complicated Problems? ---

Decompose the problem into intermediate steps and solve each before giving the final answer.

Example

"After Jane gives 2 flowers to her mom she has 10 ... then after she gives 3 to her dad she will have 7 ... so the answer is 7."

The paper shows sufficiently large LMs can generate chains of thought if demonstrations of chain-of-thought reasoning are provided as few-shot prompting.

🍷 Standard vs Chain-of-Thought Prompting ---



🍲 Benefits of Chain-of-Thought ---

- ➊ Allows models to decompose multi-step problems into intermediate steps, allocating more computation where needed.
- ➋ Provides an interpretable window into the model's reasoning process.
- ➌ Potentially applicable to any task requiring a multi-step approach.
- ➍ Can be induced simply by including chain-of-thought examples in the prompt.

(Wei, Wang, et al. 2022)

🍷 Sensitivity to Prompt Engineering ---

Chain-of-thought is robust to:

- ➊ Different annotators writing prompts.
- ➋ Annotators without machine learning background.
- ➌ Different types, ordering, and number of prompt examples.
- ➍ Different language models (LaMDA, GPT-3, PaLM).

However, prompt engineering still improved performance significantly in many cases.

🍷 Robustness: Number of Exemplars ---



🍷 Experiments & Results ---

🍷 Experimental Setup -- Benchmarks ---

Benchmarks (five math word problem datasets):

- GSM8K (Cobbe et al. 2021)
- SVAMP (Patel, Baral, and Baral 2021)
- ASDiv (Miao, Liang, and Su 2020)
- AQuA
- MAWPS (Koncel-Kedziorski et al. 2016)

Experimental Setup -- Models ---

Baseline: standard few-shot prompting (Brown et al. 2020).

Chain-of-thought prompting tested on five LLMs:

- GPT-3 (Brown et al. 2020)
- LaMDA (Thoppilan et al. 2022)
- PaLM
- UL2 20B (Tay et al. 2022)
- Codex (Chen et al. 2021)

🍷 Results -- Arithmetic Reasoning ---

Three key takeaways:

- Chain-of-thought prompting is an **emergent ability of model scale**.
- Larger performance gains for **more-complicated problems**.
- GPT-3 175B and PaLM 540B **compare favorably** to prior state of the art.



🍷 Ablation Study ---

Three variations compared against full chain-of-thought:

- Equation only
- Variable compute only
- Chain-of-thought after the answer



🍷 Robustness of Chain-of-Thought ---

Findings: chain-of-thought prompting for arithmetic reasoning is robust to:

- Annotators
- Independently-written chains of thought
- Different exemplars and exemplar orders
- Various language models
- Varying numbers of exemplars



Commonsense Reasoning ---

🍷 Commonsense & Symbolic: Example Prompts ---



🍷 Results -- Commonsense Reasoning ---

- Chain-of-thought prompting works better when the model is **sufficiently large**.



🍷 Symbolic Reasoning ---

🍷 Results -- Symbolic Reasoning (OOD) ---

- **In-domain and out-of-domain** evaluations show CoT enables length generalization.



Limitations & Discussion ---

Limitations

- CoT emulates human reasoning but does not prove the network actually "reasons."
- Emergence only at large model scales makes deployment costly.

Discussion Questions

- What reasoning tasks may not be well-suited to chain-of-thought?
- What other prompting methods might expand LM capabilities?
- Does chain-of-thought truly improve interpretability?

🍷 Can Language Models Learn from Explanations in Context? ---

(Lampinen et al. 2022)



Introduction ---

- Language models can perform new tasks by adapting to a few in-context examples (**few-shot learning**) (Brown et al. 2020).
- **Key question:** Could including few-shot *explanations* of the answers in these examples improve LM performance?

🍷 Setup: Training vs Evaluation ---



🍷 Related Work ---

- **In-context learning**: Min et al. (2022) show ground truth labels matter less than label space/distribution; Webson & Pavlick (2021) question whether models truly understand prompts.
- **Explicit instructions**: Liu et al. (2021) show task descriptions help; Wei et al. (2022) show step-by-step reasoning improves few-shot performance.
- **This paper** focuses specifically on the effect of **post-answer** explanations — distinct from chain-of-thought which places reasoning *before* the answer.

(Lampinen et al. 2022)

🍲 Research Questions ---

- Can language models benefit from explanations when learning from examples in-context?
- Do few-shot explanations help the model “understand” the task better?
- What kind of in-context learning abilities do LMs exhibit more generally?

🍲 Contributions & Main Findings ---

- Annotated **40 challenging tasks** from BIG-Bench with human-written explanations.
- Evaluated the effect of **post-answer explanations** (contrast to Wei et al.'s pre-answer chains).

Key findings

- Explanations improve LLM performance compared to matched control conditions.
- Explanations tuned on a small validation set yield even larger gains.

(Lampinen et al. 2022)

Methods: Data & Model ---

Data: 40 challenging tasks from BIG-Bench, including:

- Inferring goals from actions
- Reasoning about mathematical induction
- Reasoning about causality
- Inferring assumptions behind statements

Model: Decoder-only Transformer LMs (Gopher family: 1B, 7B, 280B parameters), evaluated with the same context window and training data.

Methods: Explanation Annotation ---

- One author annotated **15 examples per task** with human explanations (restricted to multiple-choice tasks).
- Three **control conditions** to isolate the causal effect of explanations:
 - ▶ *True non-explanations*: relevant but non-explanatory text.
 - ▶ *Other item explanations*: explanations drawn from a different task example.
 - ▶ *Scrambled explanations*: words of the explanation randomly reordered.
- **Explanation fine-tuning**: greedily select best 5-shot prompt on a 10-example validation set; also hand-tune explanation wording.

Methods: Evaluation ---

- Results modeled with **hierarchical logistic regression** to account for:
 - ▶ Task difficulty
 - ▶ Shared content between prompts
- Each effect estimated at the appropriate level of the hierarchy, quantifying the **unique added contribution** of each prompt component.

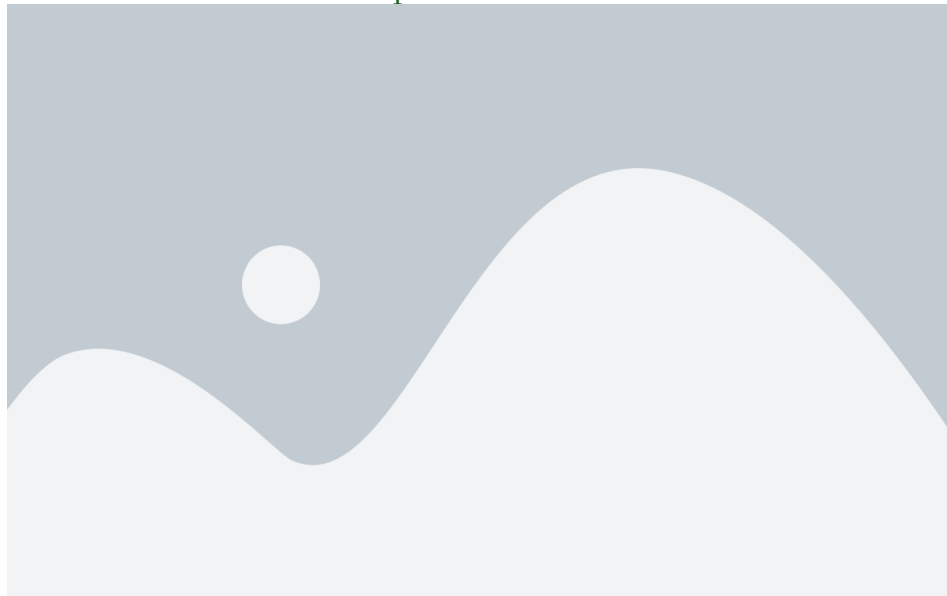
🍷 How Much Do Explanations Help? ---



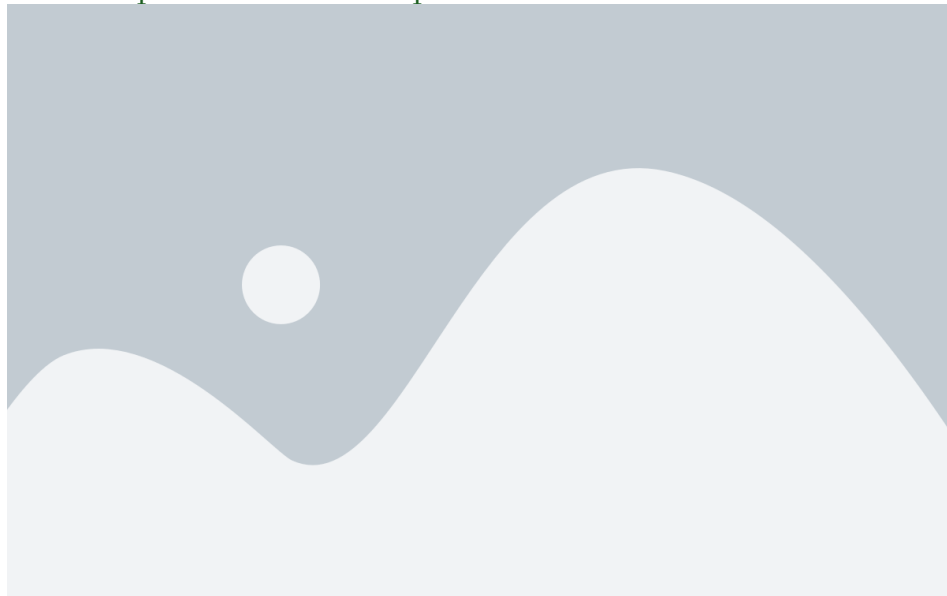
🍷 Effect Size (Gopher 280B) ---



🍷 Selection vs Untuned Explanations ---



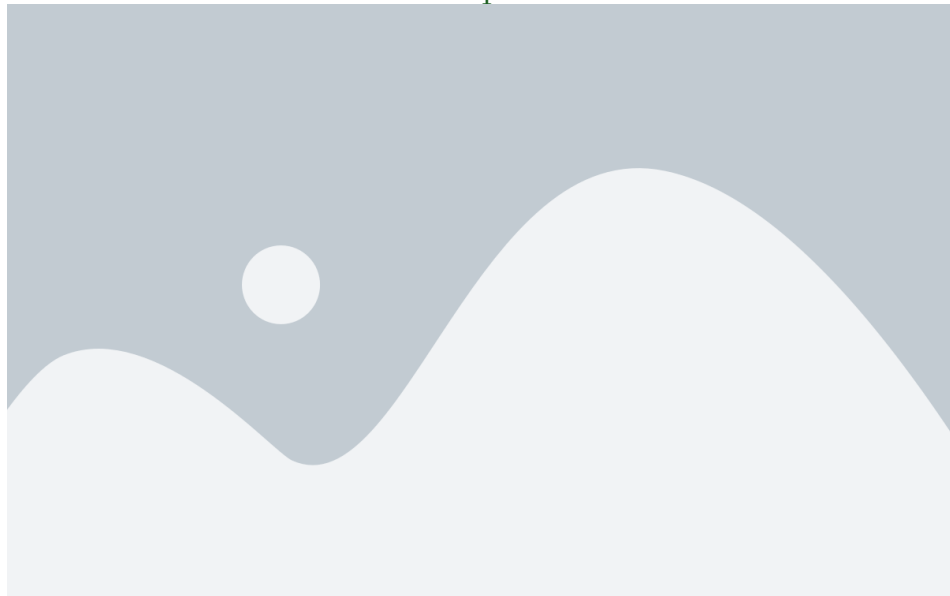
🍲 When Do Explanations Help? ---



🍷 Is Scale Necessary? ---



🍷 Not All Tasks Are Created Equal ---



Open questions

- Can explanations improve few-shot learning? **Yes, with sufficient scale and quality.**
- Why are post-answer explanations interesting? They decouple reasoning from generation and have no test-time cost.
- What do results imply about in-context learning? LMs can extract semantic content from explanations, not just surface structure.
- How do explanations relate to instructions? They appear complementary.
- How does this relate to human language processing? LMs differ — they lack broader world context.

Class Discussion ---

- Is the dataset large (~600 samples) / diverse (40 tasks) enough to support the conclusions?
- Is the degree of improvement offered by explanations practically significant?
- How does this compare to other in-context learning methods?
 - ▶ Chain-of-thought (Wei, Wang, et al. 2022)
 - ▶ Soft prompts / prompt tuning
 - ▶ Finetuning + distillation (Alpaca)
- How could explanations benefit smaller models?

Thank You!

References --- |

- Brown, Tom, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, et al. 2020. "Language Models Are Few-Shot Learners." *Advances in Neural Information Processing Systems* 33: 1877–1901.
- Chen, Mark, Jerry Tworek, Heewoo Jun, Qiming Yuan, Henrique Ponde de Oliveira Pinto, Jared Kaplan, Harri Edwards, et al. 2021. "Evaluating Large Language Models Trained on Code." *arXiv Preprint arXiv:2107.03374*.
- Cobbe, Karl, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, et al. 2021. "Training Verifiers to Solve Math Word Problems." *arXiv Preprint arXiv:2110.14168*.
- Koncel-Kedziorski, Rik, Subhro Roy, Aida Amini, Nate Kushman, and Hannaneh Hajishirzi. 2016. "MAWPS: A Math Word Problem Repository." In *Proceedings of the 2016 Conference of the North American Chapter of the Association for Computational Linguistics*, 1152–7.
- Lampinen, Andrew K, Ishita Dasgupta, Stephanie CY Chan, Kory Matthewson, Michael Henry Tessler, Antonia Creswell, James L McClelland, Jane X Wang, and Felix Hill. 2022. "Can Language Models Learn from Explanations in Context?" *arXiv Preprint arXiv:2204.02329*.

🍷 References ---] 🍷 References ---

- Ling, Wang, Dani Yogatama, Chris Dyer, and Phil Blunsom. 2017. “Program Induction by Rationale Generation: Learning to Solve and Explain Algebraic Word Problems.” *arXiv Preprint arXiv:1705.04146*.
- Miao, Shen-yun, Chao-Chun Liang, and Keh-Yih Su. 2020. “A Diverse Corpus for Evaluating and Developing English Math Word Problem Solvers.” In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, 975–84.
- Patel, Arkil, Chitta Baral, and Chitta Baral. 2021. “Are NLP Models Really Able to Solve Simple Math Word Problems?” In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics*, 2080–94.
- Tay, Yi, Mostafa Dehghani, Vinh Q Tran, Xavier Garcia, Dara Bahri, Tal Schuster, Huaixiu Steven Zheng, Neil Houlsby, and Donald Metzler. 2022. “UL2: Unifying Language Learning Paradigms.” *arXiv Preprint arXiv:2205.05131*.
- Thoppilan, Romal, Daniel De Freitas, Jamie Hall, Noam Shazeer, Apoorv Kulshreshtha, Heng-Tze Cheng, Alicia Jin, et al. 2022. “LaMDA: Language Models for Dialog Applications.” *arXiv Preprint arXiv:2201.08239*.



- Wei, Jason, Yi Tay, Rishi Bommasani, Colin Raffel, Barret Zoph, Sebastian Borgeaud, Dani Yogatama, et al. 2022. “Emergent Abilities of Large Language Models.” *Transactions on Machine Learning Research*.
- Wei, Jason, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed Chi, Quoc V Le, and Denny Zhou. 2022. “Chain-of-Thought Prompting Elicits Reasoning in Large Language Models.” *Advances in Neural Information Processing Systems* 35: 24824–37.